Practice Midterm 1

1. **Proposition:** Every positive integer n can be written as a sum of <u>distinct</u> squares of positive integers. (For example, $45 = 2^2 + 4^2 + 5^2$.)

   (a) Show that the proposition is false.

   (b) Underline and explain the mistake in the following "proof".

   Proof. We prove the proposition by strong induction on n. The base case $n = 1$ holds because $1 = 1^2$. For the inductive step, assume that it is true for all numbers between 1 and $n - 1$. If n is the square of some number, then it is also true for n. If not, $m^2 < n < (m+1)^2$ for some number $m > 0$. By inductive hypothesis, the number $n - m^2$ is a sum of distinct squares $a_1^2 + \cdots + a_k^2$. Then $n = a_1^2 + \cdots + a_k^2 + m^2$ so n is also a sum of distinct squares. $\square$

2. Alice has infinitely many \$6, \$10, and \$15 stamps.

   (a) Can she make all integer postages of \$30 and above?

   (b) What if the \$10 stamp was replaced by a \$9 stamp?

3. Prove that for every integer n there exists an integer k such that $|n^2 - 5k| \leq 1$. (**Hint:** What is n^2 mod 5?)

4. The numbers `12345678` are listed in order. In each step you can take three consecutive numbers abc and reorder them as cab, for example `2543<u>168</u>7` → `2543<u>816</u>7`. Can you ever obtain `81234567`?

Practice Midterm 2

1. Express the sentence "Any two people who are not friends have a friend in common" using quantifiers and logical operators. Use x, y, z as variables and $F(x, y)$ for "x and y are friends."

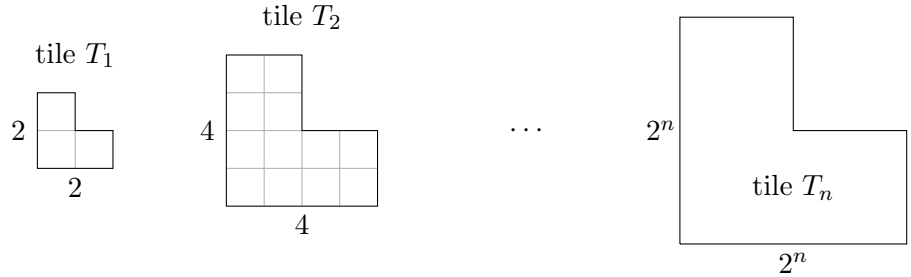
2. Show that for every integer n, if $n^3 + n$ is divisible by 3 then $2n^3 + 1$ is *not* divisible by 3.

3. True or false? Justify your answer. Specify your proof method.

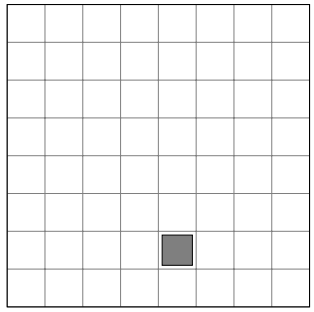
   (a) For all integers a, b, c, at least one of the three numbers $a + b$, $b + c$, $c + a$ is even.

   (b) For all integers a, b, c, at least one of the three numbers $a + b$, $b + c$, $c + a$ is odd.

4. **Claim:** For every $n \geq 1$, a $2^n \times 2^n$ board with one square removed (in any position) can be filled with tiles $T_1, \ldots, T_n$ below (one of each type).



   (a) Describe a tiling for the following board ($n = 3$ with square $(5, 2)$ missing).



(b) Prove the claim. Specify your proof method.

Practice Midterm 3

1. Underline and explain the mistake in the following "proof."

   Proposition. In every group of friends there exists a person with an even number of friends.

   Proof. By induction on the number of people n. When $n = 1$ the one person has zero friends, and zero is even. Now assume it is true for groups of n people. Let G be a group of $n + 1$ people. Take out any person from G. By inductive hypothesis the remaining group G' has someone, say Alice, with an even number of friends. Since Alice is also in G, G has a person with an even number of friends. $\square$

2. Prove that for every positive integer n, $\gcd(n^2 + n + 1, n + 1) = 1$.
   (**Hint:** Use the connection between gcd and combinations.)

3. For which nonzero integers n is the number $\frac{\sqrt{2}}{n} - \frac{n}{\sqrt{2}}$ rational? Justify your answer.

4. Bob has 32 *b*lue, 33 *r*ed, and 34 *g*reen balls. At every turn he takes out two balls and replaces them with two different balls by the following replacement rule:

$$bg \to rr \quad gr \to bb \quad rb \to gg \quad rr \to bg \quad bb \to gr \quad gg \to rb.$$

   (a) Formulate this game as a state machine. Describe the states, start state, and transitions mathematically.

   (b) Can Bob obtain 99 balls of the same color? Justify your answer.
   (**Hint:** Look at the difference between the number of red and blue balls.)

Practice Midterm 4

1. Are the propositions "Every two people have a common friend" and "Every person has at least two friends" logically equivalent? Justify your answer.

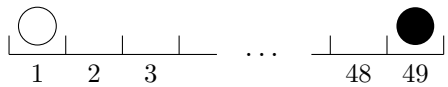
2. Show that for every positive real number x, at least one of the numbers $\sqrt{x} + 1$ and $\sqrt{2} \cdot x$ is irrational.

3. Bob is waiting for a secret message from Alice. He publishes RSA modulus $n = 21$ and public key $e = 5$.

   (a) Alice's message is $m = 8$. What is the ciphertext that she sends out? Show your calculations.

   (b) Alice sends another ciphertext $c = 2$ and this one is intercepted by Eve. What was Alice's message?

4. A long ledge is divided into slots numbered from 1 to 49.



   A white ball and a black ball are placed in the first and last slot, respectively. In every step one of the balls is moved 5 slots to the left *or* 10 slots to the right of its current position.

(a) Formulate this process as a state machine. Describe the states, start state, and transitions.

(b) *Invariant:* The slot numbers of the black and white balls differ by ____ modulo ____ . Fill in the blanks and provide a proof of invariance.

(c) Can the two balls ever occupy adjacent slots? Justify your answer.

1. Are the propositions "Every two people have a common friend" and "Every person has at least two friends" logically equivalent? Justify your answer.

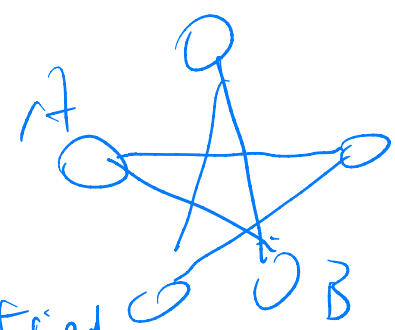
If there's only one person, then the statements are T and F respectively so they are not equivalent.

Let 5 people be denoted by circle and their friendship be denoted by lines

Then the first statement is T

but the second statement is F

as A and B have no common friend



2. Show that for every positive real number x, at least one of the numbers $\sqrt{x}+1$ and $\sqrt{2}\cdot x$ is irrational.

Let $b=\sqrt{x}$

$\sqrt{x}=b+1$, $\sqrt{2}\,x = b^2\sqrt{2}$.

If $b+1$ is rational then b is rational,

then $b^2\sqrt{2}$ is irrational

If $b^2\sqrt{2}$ is rational, then b^2 is irrational

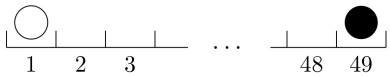
then b must be irrational. then $b+1$ must be irrational

(a) Formulate this process as a state machine. Describe the states, start state, and transitions.

(b) *Invariant:* The slot numbers of the black and white balls differ by ____ modulo ____ . Fill in the blanks and provide a proof of invariance.

(c) Can the two balls ever occupy adjacent slots? Justify your answer.

4. A long ledge is divided into slots numbered from 1 to 49.



A white ball and a black ball are placed in the first and last slot, respectively. In every step one of the balls is moved 5 slots to the left *or* 10 slots to the right of its current position.

(a) Formulate this process as a state machine. Describe the states, start state, and transitions.

(b) *Invariant:* The slot numbers of the black and white balls differ by ____ modulo ____ . Fill in the blanks and provide a proof of invariance.

(c) Can the two balls ever occupy adjacent slots? Justify your answer.

(a) state is (x, y) where x is the position of the white ball, y is the position of the black ball

start state is $(1, 49)$

state transition is

$(x-5, y), (x, y-5), (x+10, y), (x, y+10) \in \{1, \ldots, 49\}^2$

(b) The four transitions gives $x-y$

$-5, +5, +10, -10$ respectively.

which means $x-y \pmod 5$ is not changing.

$x-y = -48 \pmod 5$

as an invariant.

(c) $x-y \pmod 5$ cannot be $+1$ or 1 so they cannot be adjacent.